HACKATHON 7 · SDG 15.8 AND 15.5

HORNET CHECK NL

Is that the invasive Asian hornet? Take a photo and find out what to do next.

Minhyeok (Min) Sung · Sam Kempers

THE PROBLEM

An invasive bee hunter is spreading fast

2,500+

Asian hornet nests reported in the Netherlands in **2024**, via waarneming.nl

2017: the first Dutch nest

5 to 6× more nests every year

Hunts **honeybees and wild pollinators** in front of their nests

Source: T. Zeegers, EIS Kenniscentrum Insecten, “Kijk op Exoten” nr. 50, September 2025

WHY IT'S HARD

Five insects that look alike

INVASIVE

Asian hornet

Vespa velutina

Dark body, one orange band, yellow leg tips

NATIVE

European hornet

Vespa crabro

Bigger, brown-red and yellow

NATIVE

Wasp

Vespula spp.

Smaller, bright black and yellow stripes

NATIVE

Hornet hoverfly

Volucella zonaria

A harmless fly: big eyes, no sting

NATIVE

Honeybee

Apis mellifera

The victim: brown, hairy, smaller

Most people can't tell them apart. The result: **missed reports, false alarms, and useful native insects killed.**

OUR USERS

Beekeepers and Bee enthusiasts in the Netherlands

Who it’s for

- People that want to check what species the ‘bee’ is.
- People who don't have an expertise within insects.

Who it’s not for

- People who have their expertise within insects
- Use outside of the Netherlands

HOW IT WORKS

From photo to action in four steps

1. Photo

Upload or take one with the camera

→

2. Roboflow

Our model gives a score for each of the 5 insects

→

3. Rules

Our code checks how sure the model is

→

4. Action

Species, advice, and a report button

70% or more

Clear answer. Asian hornet: warning plus “report it”

45 to 70%

“Probably X”, plus the second guess

Under 45%

“I don’t know. Report it if in doubt”

THE DATA

Our own Dutch dataset, scraped from iNaturalist

1,215	539	887	121
photos downloaded	different photographers	training photos after cleaning	never-seen test photos

- **Research-grade only:** the species was confirmed by other users
- **Netherlands only:** each location checked against the nearest town
- **Trained in Roboflow:** classification, DINOv3 Base, 224×224 px, 3× augmentation

V1 → V2: WHAT WE CHANGED

REMOVING 207 BAD PHOTOS HALVED THE CONFIDENT MISTAKES

23% → 13%

In v1, **22% of the Asian hornet photos showed no usable insect**: nests, traps, phone screenshots, people. We checked all 1,094 training photos by eye.

RESULTS

v2 is far less often confidently wrong

Test set	v1	v2
When it says “Asian hornet”, it’s right (precision)	57%	82%
Asian hornet F1	58.5%	66.7%
Real Asian hornets found (recall)	60%	56%
Wasp called “Asian hornet” (false alarm)	5	1
Asian hornet called “European hornet” (missed nest)	3	1
Confidently wrong answers	23%	13%

It still misses about half of Asian hornets, but mostly by being unsure, not wrong. That’s why the app says “report it if in doubt”.

Roboflow test splits: v1 109 photos, v2 94 photos (about 16–20 Asian hornets each), so the comparison is indicative

LIVE DEMO

LET’S TRY IT

1

An Asian hornet

Clear warning, advice and the report button

2

A European hornet

Native: leave it alone, and how to tell them apart

3

A difficult photo

The app admits it isn’t sure

Photos from our never-seen test set · App: [streamlit link]

ETHICS

What can go wrong, and what we did

Risk	What could happen	What we did
Safety Issue	People get closer to insects that could harm: Stings, dangerous with allergies.	The app says “Keep a few meters away”.
Disturbing animal protection	People could disturb the very animals that are under protection.	The app says “Take the photo from a distance, and never touch or remove a nest”
Over-trust	People stop looking for themselves	Confidence always shown; experts confirm reports

WHAT WE LEARNED

Three lessons

01

Look at your data

Removing bad photos helped more than any setting. “Research-grade” doesn’t mean “good training photo”.

02

Doubt is a feature

An app that says “I’m not sure” is safer than one that always gives an answer.

03

The right metric matters

Overall F1 barely moved, but the dangerous mistakes dropped. Ask which errors hurt users.

NEXT STEPS

- More Asian hornet photos, especially flying and at the hive
- A user test with beekeepers
- Export the model so it also works offline

THANK YOU

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GitHub: [repo link] · App: [streamlit link]